

```

import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
print("Upload an image containing one or more faces")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image
img = cv2.imread(filename)
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# -----
# Low-Level Vision
# Image Enhancement
# -----

enhanced = cv2.equalizeHist(gray)

# -----
# Mid-Level Vision
# Face Detection
# -----

face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + 'haarcascade_frontalface_default.xml'
)

faces = face_cascade.detectMultiScale(
    enhanced,
    scaleFactor=1.1,
    minNeighbors=5
)

# Draw rectangles
output = img_rgb.copy()

for (x, y, w, h) in faces:
    cv2.rectangle(output, (x, y), (x+w, y+h), (0,255,0), 2)

# -----
# Display
# -----

plt.figure(figsize=(12,6))

plt.subplot(1,2,1)
plt.imshow(img_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(output)
plt.title("Detected Faces")
plt.axis("off")

plt.show()

# -----
# High-Level Vision
# -----

```

```
print("Computer Vision Levels in Facial Recognition\n")
```

```
print("1. Low-Level Vision")
print("  - Image acquisition")
print("  - Histogram Equalization")
print("  - Noise reduction")
```

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```
print("\n2. Mid-Level Vision")
print("  - Face detection")
print("  - Feature extraction")
```

```
print("\n3. High-Level Vision")
print("  - Face recognition")
print("  - Identity verification")
print("  - Decision making")
```

```
print(f"\nFaces Detected: {len(faces)}")
```

Upload an image containing one or more faces

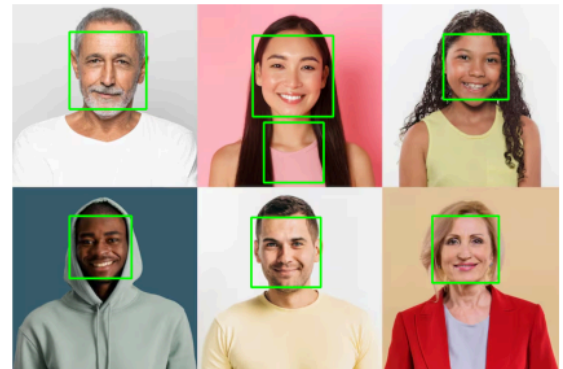
people-colla...888275.avif

people-collage-design_23-2148888275.avif(image/avif) - 25074 bytes, last modified: 14/07/2026 - 100% done
 Saving people-collage-design_23-2148888275.avif to people-collage-design_23-2148888275.avif

Original Image



Detected Faces



Computer Vision Levels in Facial Recognition

1. Low-Level Vision
 - Image acquisition
 - Histogram Equalization
 - Noise reduction
2. Mid-Level Vision
 - Face detection
 - Feature extraction
3. High-Level Vision
 - Face recognition
 - Identity verification
 - Decision making

Faces Detected: 7

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